

PROFILE

Applied computer vision engineer with a geoinformatics background. I run two detection pipelines: parking occupancy from a camera stream, operating unattended from frame to database, and road sign detection from mobile mapping in the CEDA Maps R&D team. The work covers the whole chain — annotation, dataset engineering against data leakage and class imbalance, training, inference and deployment. I use agentic development tooling daily and encoded the domain rules of spatial data work into it as my own extension.

EXPERIENCE

GIS & Computer Vision Analyst (contractor) — CEDA Maps

June 2026 – present

Brno, remote

- Building a GeoAI pipeline for automatic detection and classification of road signs from mobile mapping — from data annotation through model training to production deployment
- Localising signs without precise GNSS: multi-view triangulation of observations from dozens of passes, least-squares refinement and fusion into a single entity; monocular distance estimation (Depth Anything V2, MoGe-2) as an independent source, validated by a null test and scenario synthetic data
- Calibration of a panoramic camera rig via COLMAP SfM — extrinsics and intrinsics for lifting detections into real-world space
- Detecting sign panels directly in the LiDAR point cloud (RIEGL) and cross-validating them against image detections
- Dataset engineering: per-track splits against data leakage, deduplication of sequential frames and class-imbalance control over tens of thousands of crops; synthetic scene augmentation for weather and lighting
- Python, PyTorch, Ultralytics YOLOv8/OBB, SAHI, ONNX Runtime, OpenCV, Open3D, laspy, COLMAP, CVAT, PostGIS, GeoPandas, Shapely

Co-founder and technical lead — VečerkaPlus

April 2026 – present

Frýdek-Místek, Czechia · vecerkaplus.cz

- Co-owner and technical lead of a late-night drinks and snacks delivery service in Frýdek-Místek — the entire web platform built from scratch on React, Supabase and Vercel
- Delivery zone system built on spatial analysis: 1,093 grid points via the Google Distance Matrix API, so price and availability follow real driving distance rather than straight-line distance
- Supabase backend: PostgreSQL with PL/pgSQL triggers and RLS policies, Vercel Edge Functions, automated notifications via Resend, a Telegram bot and Twilio
- Running the application in production, incidents reported by customers included — fixing a crash of the ordering flow on Safari/iOS; a redesign of the courier portal and verification of the whole order through automated browser runs
- Alongside development I run operations — purchasing, courier logistics and growth strategy; average revenue of CZK 452 per order at a 36.5 % margin

Web Developer & Administrator — Palacký University Olomouc

May 2023 – present

Olomouc, Czechia

- Running olomouckymajales.cz and meetup.upol.cz — 199,800 visitors and 452,000 page views in total
- Requested changes delivered within 24 hours and uninterrupted operation during key events; design and functional changes deployed to production with no downtime
- WordPress, YOOftheme Builder

GIS & Remote Sensing Analyst — Skymaps s.r.o.

February 2025 – September 2025

Czechia

- Processing multispectral satellite imagery into thematic maps of soil potential
- Analysis and interpretation of remote sensing data: Sentinel in SNAP, GDAL, EO Browser; Python automation from batch image processing to final map outputs

Drone Operator — Aerial imaging (freelance)

2024 – 2025

Ostrava, remote

- Commissioned drone footage for corporate marketing materials; planning and flying missions in compliance with EASA A1/A3 rules (DJI Mini 3 Pro)

SELECTED PROJECTS

Parking Occupancy Detection from a Camera Stream

2025 – 2026

A system deriving parking occupancy from site cameras that runs unattended: a scheduler triggers capture, detection and the write to PostgreSQL, with follow-up scripts backfilling gaps. Inference combines several detection models with custom overlap merging, and frames are georeferenced into the Czech national grid. A separate branch reads number plates and validates the OCR output against a Czech plate mask, so nonsense never reaches the data. The dashboard API is a deliberately separate FastAPI service, keeping GPU dependencies off the API server.

Python · Ultralytics YOLO · PyTorch · OpenCV · FastAPI · PostgreSQL · rasterio · pyproj · pytest

Own Tooling for Agentic Development

2026

The *geospatial-engineer* extension — a specification and seven reference documents — encodes the domain rules of spatial data work: coordinate systems, area-of-interest boundaries and the silent failures of raster processing that otherwise have to be restated in every project. Alongside it, a portable development environment configuration with its own memory system, and automated AI code review in CI via GitHub Actions. A pull request extending process sampling beyond Linux (psutil) is open against the open-source tool *aw-watcher-git*.

Claude Code · Markdown · PowerShell · GitHub Actions

LLM-Generated Map Evaluation

2026 – in progress

github.com/MetrPikeska/ai-generated-map-evaluation

Master's thesis: a systematic comparison of maps generated by five large language models (ChatGPT, Claude, Gemini, Mistral, Perplexity) against a reference workflow in ArcGIS Pro, using object-based metrics and cartographic analysis. A Python batch pipeline, with results in an interactive Leaflet map.

Python · GeoPandas · Leaflet · GIS metrics

Vehicle Detection on Orthophotos

2025 – 2026

github.com/MetrPikeska/vehicle-detection-ortophoto

Vehicle detection on large-area orthophotos of Olomouc, fully integrated into a GIS workflow. YOLOv8-OBb with SAHI enables inference on images larger than the GPU can hold; DBSCAN clustering of detections and Voronoi-based address assignment in GeoPandas follow. An end-to-end pipeline from CVAT annotation through training and georeferencing to spatial analysis. Presented at the international POGEO 2026 conference.

Python · YOLOv8-OBb · SAHI · GeoPandas · Shapely · CUDA

VečerkaPlus

2026

vecerkaplus.cz

A production ordering platform for late-night delivery. React 19, Vite and TypeScript on the front end with a bespoke design system and no third-party UI framework; a Supabase backend with PostgreSQL, PL/pgSQL triggers and RLS. The delivery zone is computed from 1,093 grid points via the Google Distance Matrix API.

React · TypeScript · Supabase · PostgreSQL · Google Maps API · Vercel

GEOTE Klima

2025

petrmikeska.cz/geote

A full-stack Web GIS platform for visualising and querying climate and environmental data. PostGIS with spatial indexes behind a Flask REST API, and a Leaflet front end with dynamic filtering directly on the map.

PostGIS · PostgreSQL · Flask · REST API · Leaflet · JavaScript

SKILLS

Languages & data-bases	Python · SQL / PL-pgSQL · JavaScript / TypeScript · R · Bash · C# · PostgreSQL / PostGIS · MySQL · Supabase
Computer vision & machine learning	PyTorch · Ultralytics YOLOv8 / OBb · SAHI · OpenCV · torchvision · Hugging Face Transformers · Depth Anything V2 · MoGe-2 · scikit-learn · ONNX Runtime · OCR / ANPR · CVAT · CUDA
GIS & spatial data	QGIS · ArcGIS Pro / ArcPy · GDAL/OGR · GeoServer · MapServer · SNAP · GeoPandas · Shapely · pyproj · Rasterio · H3 · Folium · OGC standards · EPSG 5514 / 3045 / 4326
LiDAR, 3D & photogrammetry	Point cloud processing (RIEGL) · Open3D · laspy · CloudCompare · Agisoft Metashape · COLMAP (SfM) · multi-view triangulation · camera rig calibration · DTM/DSM · Blender · MeshLab
Web & backend	React · Vite · Leaflet · FastAPI · Flask · Node.js / Express · REST API · WebSocket · HTML/CSS · WordPress · Vercel
Tooling & operations	Git / GitHub · Linux / Ubuntu / WSL · Docker · pytest · NumPy · Pandas · SciPy · Matplotlib
AI-assisted development	Claude Code — custom extensions, project rules and memory, orchestration of parallel agents · MCP clients (Supabase, Vercel, browser automation) · GitHub Copilot · AI code review in CI (GitHub Actions)
Domain & standards	TP 65 (Czech road sign standard) · HD maps · mobile mapping systems · GNSS and NMEA · TomTom / StreetNet · RÚIAN · Sentinel / remote sensing · INSPIRE

EDUCATION

MSc — Geoinformatics and Cartography (in progress)

2025 – 2027

Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics

- Focus: advanced spatial data processing, web cartography, remote sensing and image processing
- Master's thesis (in progress): *Evaluating LLM-generated maps against a traditional GIS workflow*
- Erasmus+ — study period in Greece, October 2026 – February 2027
- Drones, laser scanning, 3D printing and virtual reality

BSc — Geoinformatics and Cartography

2022 – 2025

Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics

- Bachelor's thesis: *A user GIS toolbox for assessing the accessibility of parks and green spaces for urban residents* — grade B

Maturita — Programming and Application Development / Internet of Things

2018 – 2022

Secondary School of Information Technology · Frýdek-Místek, Czechia

- Programming, web technologies, computer networks, databases and the fundamentals of electronics and IoT

CERTIFICATIONS

October 2024	Remote pilot certificate, category A1/A3 · Civil Aviation Authority of the Czech Republic / EASA
May 2023	Python for Everyone · Esri
April 2026	ISSonVIS 2026 — International Spring School on Visualization · Dept. of Geoinformatics, Palacký University Olomouc

LANGUAGES

Czech (native) · English (B2)